8.

On Being Round

Apart from crystals and broken rocks, not much else in the cosmos naturally comes with sharp angles. While many objects have peculiar shapes, the list of round things is practically endless and ranges from simple soap bubbles to the entire observable universe. Of all shapes, spheres are favored by the action of simple physical laws. So prevalent is this tendency that often we assume something is spherical in a mental experiment just to glean basic insight even when we know that the object is decidedly non-spherical. In short, if you do not understand the spherical case, then you cannot claim to understand the basic physics of the object.

Spheres in nature are made by forces, such as surface tension, that want to make objects smaller in all directions. The surface tension of the liquid that makes a soap bubble squeezes air in all directions. It will, within moments of being formed, enclose the volume of air using the least possible surface area. This makes the strongest possible bubble because the soapy film will not have to be spread any thinner than is absolutely necessary. Using freshman-level calculus you can show that the one and only shape that has the smallest surface area for an enclosed volume is a perfect sphere. In fact, billions of dollars could be saved annually on packaging materials if all shipping boxes and all packages of food in the supermarket were spheres. For example, the contents of a super-jumbo box of Cheerios would fit easily into a spherical carton with a four-and-a-half-inch radius. But practical matters prevail—nobody wants to chase packaged food down the aisle after it rolls off the shelves.

On Earth, one way to make ball bearings is to machine them, or drop molten metal in pre-measured amounts into the top of a long shaft. The